Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)		
Lab exp 11	**Date: 23/10/2026**	**Enrolment No:92400133079**

CODE

```
import cv2
import numpy as np
from google.colab.patches import cv2_imshow

# 1. Load the image in grayscale
image = cv2.imread('/content/IMG_20260315_113925.jpg.jpeg', 0)

# Check whether the image was loaded successfully
if image is None:
    raise FileNotFoundError("Image not found. Check the image path.")

# 2. Define the structuring element (kernel)
kernel = np.ones((5, 5), np.uint8)

# 3. Perform Erosion
erosion = cv2.erode(image, kernel, iterations=1)

# 4. Perform Dilation
dilation = cv2.dilate(image, kernel, iterations=1)

# 5. Perform Opening (Erosion followed by Dilation)
opening = cv2.morphologyEx(image, cv2.MORPH_OPEN, kernel)

# 6. Perform Closing (Dilation followed by Erosion)
closing = cv2.morphologyEx(image, cv2.MORPH_CLOSE, kernel)

# 7. Display the original image and processed results
print("Original Image")
cv2_imshow(image)

print("Erosion")
cv2_imshow(erosion)

print("Dilation")
cv2_imshow(dilation)

print("Opening")
cv2_imshow(opening)
```

```
print("Closing")
cv2_imshow(closing)

# 8. Close OpenCV windows (not required in Google Colab)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

OUTPUT



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GITHUB : https://github.com/yugmakwana99/DSIP.git

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